

ENGR 1451/2451: Exploratory Data Science

Accuracy of Averages and Confidence Intervals for Averages

Prof. Paul W. Leu
University of Pittsburgh

Fall 2026

Today: From SE to Confidence Intervals

- Review how sample averages vary across random samples.
- Estimate the SE using the sample SD when σ is unknown.
- Use Student's t -distribution to construct confidence intervals.
- Interpret confidence intervals and understand their limitations.

Brief Recap: Standard Error of an Average

Known population: fair die

For one roll of a fair six-sided die,

$$\mu = 3.5, \quad \sigma = \sqrt{35/12} \approx 1.708.$$

Average of n independent rolls

$$E(\bar{X}) = \mu, \quad SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Example: $n = 36$

$$SE(\bar{X}) = \frac{1.708}{\sqrt{36}} \approx 0.285.$$

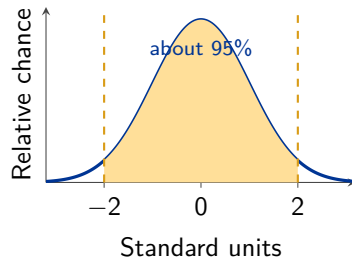
- σ describes individual rolls.
- $SE(\bar{X})$ describes sample averages.

Why the Normal Curve Can Be Used

Central-limit idea

For many independent rolls, the probability histogram for the **sample average** is close to the normal curve after converting to standard units.

- A single roll is uniform on 1, 2, 3, 4, 5, 6.
- The average of many rolls is much more bell-shaped.
- The normal curve approximates the sampling distribution of $\bar{X}$.



First Pass: Confidence Interval Idea

Normal-curve rule of thumb

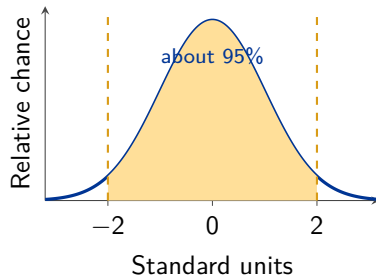
If the sampling distribution of $\bar{X}$ is approximately normal,

$$\text{estimate} \pm 2 \times \text{SE}$$

gives an approximate 95% confidence interval.

Large-sample shortcut

$$95\% \text{ CI} \approx \bar{x} \pm 2\widehat{\text{SE}}$$



Real Data: The Population Mean Is Unknown

Known-population reasoning

For the die, μ and σ are known.

$\bar{X}$ is centered at μ , $SE(\bar{X}) = \sigma/\sqrt{n}$.

We can ask: how often will $\bar{X}$ land near μ ?

Real-data reasoning

In real data, μ and σ are usually unknown.

$\bar{x}$ estimates μ , $\widehat{SE}(\bar{X}) = s/\sqrt{n}$.

Now we ask: how accurate is the estimate $\bar{x}$?

Use the estimated SE to put a margin of error around a sample average.

There Is No Standard $\pm$ Convention

The symbol $\pm$ does not have one universal statistical meaning.

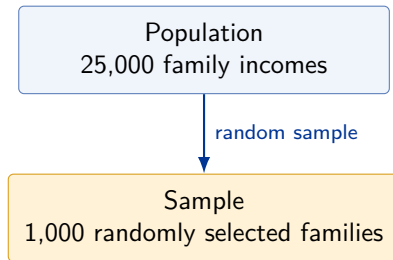
- mean $\pm$ sample standard deviation (SD)
- mean $\pm$ standard error of the mean (SE)
- mean $\pm$ margin of error for a confidence interval

Always state what the $\pm$ value represents.

Real Survey Example: Average Family Income

Study setup

- Town has 25,000 families.
- Simple random sample of 1,000 families.
- Sample total income: \$62,396,714.
- Sample average: about \$62,400.
- Sample SD: about \$53,000.



Use $\bar{x} = \$62,400$
to estimate the population average

Estimate the SE for the Sample Average

Use the sample SD because σ is unknown

$$\widehat{SE}(\bar{X}) = \frac{s}{\sqrt{n}} = \frac{\$53,000}{\sqrt{1000}} \approx \$1,700.$$

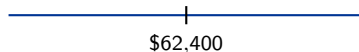
- \$53,000: spread among individual family incomes.
- \$1,700: estimated sampling variability of the average.

Point estimate and estimated SE

$$\bar{x} \approx \$62,400$$

$$\widehat{SE}(\bar{X}) \approx \$1,700$$

one estimated SE each way



Not yet a 95% confidence interval

Freedman, Pisani, and Purves, *Statistics*, 4th ed. (2007), p. 416..

Why Student's t Shows Up

If σ were known

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1).$$

In real data, σ is unknown

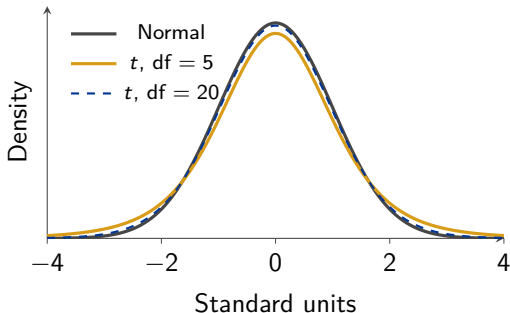
$$\widehat{SE}(\bar{X}) = \frac{s}{\sqrt{n}}.$$

Now the standardizer also varies from sample to sample:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}.$$

Estimating the SE adds uncertainty. Student's t accounts for this by using a slightly wider curve.

Student's t Distribution



Main features

- Heavier tails than the normal curve.
- Shape depends on degrees of freedom.
- As df increases, t approaches the normal curve.

For one sample mean

$$df = n - 1.$$

Degrees of Freedom and Critical Values

95% critical value

Sample size n	df	95% t^*
5	4	2.776
10	9	2.262
20	19	2.093
30	29	2.045
100	99	1.984
Large	–	≈ 1.96

Why $df = n - 1$?

After the sample mean $\bar{x}$ is estimated, only $n - 1$ deviations from the mean can vary freely.

Smaller samples require larger critical values and wider 95% confidence intervals.

Confidence Interval for a Population Mean

When the population SD σ is unknown

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

What the terms mean

- $\bar{x}$: sample mean
- $s/\sqrt{n}$: estimated SE
- t^* : critical value determined by confidence level and $df = n - 1$

Steps

1. Calculate $\bar{x}$, s , and n .
2. Estimate $SE = s/\sqrt{n}$.
3. Find t^* using $df = n - 1$.
4. Margin of error = $t^* \times SE$.

For a 95% CI and large n , $t^* \approx 1.96 \approx 2$.

Confidence Level and α

α (**significance level**): the probability outside the central confidence region of a reference distribution.

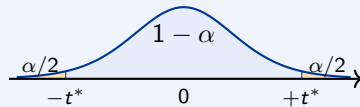
For a two-sided confidence interval

$$\text{Confidence level} = 1 - \alpha$$

Each tail contains probability $\alpha/2$.

Confidence	α	Each tail
90%	0.10	0.05
95%	0.05	0.025
99%	0.01	0.005

Middle 95%: t with $df = 9$



Normal and Student's t in Python

```
from scipy import stats
```

Standard normal

```
model = stats.norm()
```

Student's t

```
model = stats.t(df=n-1)
```

What to calculate

Python method

Density at x

`model.pdf(x)`

Area to left/right of x

`model.cdf(x)` / `model.sf(x)`

Area between a and b

`model.cdf(b) - model.cdf(a)`

Cutoff with left/right-tail area p

`model.ppf(p)` / `model.isf(p)`

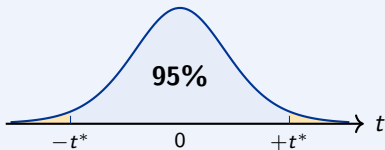
Inverse pairs: `cdf` $\leftrightarrow$ `ppf` `sf` $\leftrightarrow$ `isf`

For a general normal distribution: `stats.norm(loc=mean, scale=sd)`

Finding the 95% Critical Value

Suppose σ is unknown and a sample has $n = 10$.

Middle 95% of the t -distribution



Each tail contains 2.5% of the area.

Questions

1. What are α and df ?
2. What area is to the left of $+t^*$?
3. Which SciPy method finds a cutoff from an area?

Complete the Python call

```
stats.t.____(p, df=n-1)
```

Python: 95% Confidence Interval

Estimate the SE and calculate the interval

```
from scipy import stats

alpha = 0.05
n = len(data)
xbar = data.mean()
se = stats.sem(data)

t_star = stats.t.ppf(1 - alpha / 2, df=n - 1)
ci = (xbar - t_star * se, xbar + t_star * se)
```

Alternatively, calculate the interval directly

```
ci = stats.t.interval(1 - alpha, df=n - 1,
                     loc=xbar, scale=se)
```

Both methods return the same confidence interval.

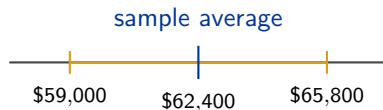
From Standard Error to Confidence Interval

95% confidence interval for this large sample

Here $n = 1000$, so $t_{999}^* \approx 1.96 \approx 2$. Go about **2 estimated SEs** either way from the sample average.

$$\$62,400 \pm 2(\$1,700) = \$62,400 \pm \$3,400.$$

$$\$59,000 \text{ to } \$65,800.$$



Large-sample shortcut

$$95\% \text{ CI} \approx \bar{x} \pm 2\widehat{SE}$$

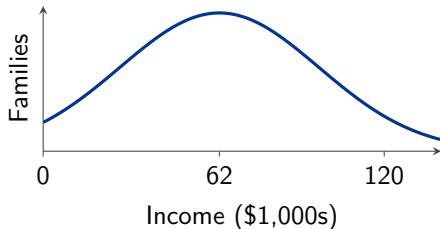
- The interval estimates the **population average**.
- It is not an interval for typical individual incomes.

Freedman, Pisani, and Purves, *Statistics*, 4th ed. (2007), pp. 416-417..

SD and SE Answer Different Questions

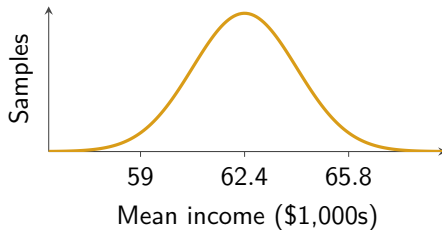
Sample SD: \$53,000

Variation among individual incomes in the sample.



SE: \$1,700

Typical sampling variability of the sample mean.



Illustrative curves, not histograms of the observed data; real income distributions may be skewed.

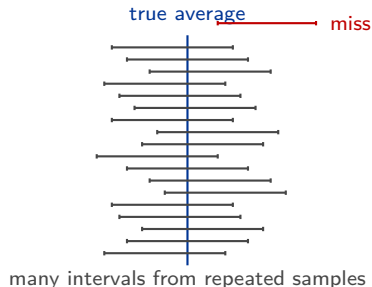
Freedman, Pisani, and Purves, *Statistics*, 4th ed. (2007), p. 417..

What Does 95% Confidence Mean?

Important interpretation

The population average is fixed. The interval changes from sample to sample.

- In about 95% of all random samples, the interval covers the population average.
- In about 5%, it misses.
- The chance is in the sampling procedure, not in the fixed population average.



Freedman, Pisani, and Purves, *Statistics*, 4th ed. (2007), p. 417..

Small-Sample Example: Why Use Student's t ?

A random sample of $n = 10$ printed parts has mean tensile strength $\bar{x} = 22.3$ MPa and sample SD $s = 4.5$ MPa. Assume independent measurements from an approximately normal population.

Estimate the SE

$$\widehat{SE} = \frac{4.5}{\sqrt{10}} \approx 1.423 \text{ MPa}$$
$$df = n - 1 = 9$$

95% critical values

$$t^* = 2.262, \quad z^* = 1.960$$

Student's t : correct method

$$22.3 \pm 2.262(1.423)$$

$$19.08 \text{ to } 25.52 \text{ MPa}$$

Normal approximation

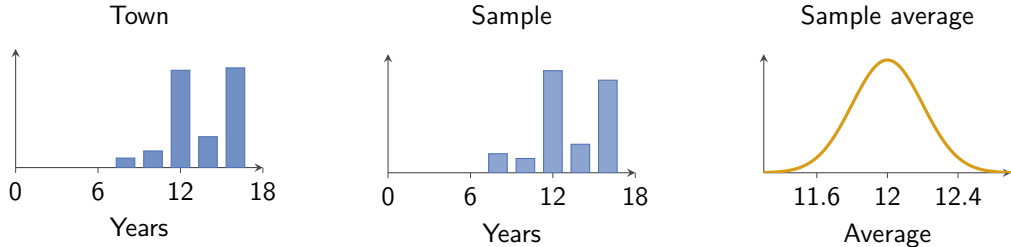
$$22.3 \pm 1.960(1.423)$$

$$19.51 \text{ to } 25.09 \text{ MPa}$$

At small n , t gives a wider interval because σ is unknown.

The Data Do Not Need to Look Normal

Illustrative example: averages across samples can be bell-shaped even when individual measurements are not.



Lesson

The normal curve is used for the **probability histogram of the sample average**, not necessarily for the histogram of the raw data.

When Can We Trust a Confidence Interval?

The usual confidence interval for a population mean is

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

Conditions

1. **Random sampling:** The sample is randomly selected from the population of interest.
2. **Independent observations:** Measurements do not systematically depend on one another.
3. **Approximately normal sample mean:** The population is approximately normal, or the sample is sufficiently large for the CLT to apply.

Important limitation

Confidence intervals account for **random sampling error**, but they do not correct **systematic bias**.

Quick Check

1. A sample average is 71.3 and the sample SD is 2.3 from $n = 500$ observations. What is the estimated SE of the average?
2. Does the interval $71.3 \pm 2(0.1)$ describe individual observations or the population average?
3. If the sample size is multiplied by 4, what happens to the SE of the average?
4. Why is it wrong to use these formulas for a convenience sample?

Summary

- The sample average estimates the population average.
- The estimate is off by chance error; the SE estimates the likely size of that error.
- For independent observations (or when dependence from sampling without replacement is negligible),

$$\widehat{SE}(\bar{X}) = \frac{s}{\sqrt{n}}.$$

- A one-sample 95% confidence interval for a population mean uses

$$\bar{x} \pm t_{n-1}^* \frac{s}{\sqrt{n}}.$$

- For large samples, $t^* \approx 1.96 \approx 2$, so the shortcut is $\bar{x} \pm 2\widehat{SE}(\bar{X})$.
- Confidence intervals require a valid sampling method; they do not fix bias or convenience sampling.